

Pricing Capability: A Framework for Valuing Proprietary Data Assets in the Age of AI

Capability Alpha, Price-Implied Uplift, and the Spirit Aviation Data Auction

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Abstract

Companies increasingly use proprietary enterprise data to train, adapt, and evaluate AI models, yet there is no established way to price workflow-linked data whose value depends on the capability it adds to a model or system. Google LLC’s selected \$10 million bid for Spirit Aviation Holdings’ deidentified operational data and software provides a rare public case in which this problem can be examined. Existing research already links information and predictive improvement to economic value, while adjacent work studies intangible assets, AI training-data markets, process data, and the contribution of data to model performance. This paper focuses on the transaction-level problem of combining these pieces for proprietary enterprise corpora used in current AI training and evaluation: measuring the marginal performance contribution of a dataset under controlled evaluation, mapping that contribution into buyer-specific economics, and interpreting an observed bid under explicit rights, useful-life, and market-structure assumptions.

We use *Capability Alpha* to denote the causal improvement in economically relevant model or system performance attributable to a proprietary dataset under controlled evaluation. The measured gain is translated into buyer-specific use value through economic exposure, the mapping from performance to commercial outcomes, the expected duration of the gain, usable rights, and acquisition and integration costs. We also define *Price-Implied Uplift* (PIU), a specialization of reverse discounted-cash-flow analysis that recovers the persistent revenue-equivalent uplift required to justify an observed price under a stated revenue-base scenario.

Applied to Spirit, an illustrative scenario in which products or workflows representing 5% of annual Google Cloud revenue form the relevant revenue base implies a three-year PIU of about 0.24% at the selected \$10 million bid. Across 1%–10% revenue-scope assumptions, PIU ranges from about 1.18% to 0.12%. A separate forward sensitivity grid shows that assumptions of a 1% improvement in airline economics and 0.5%–2.0% Google capture of the resulting value produce a present-value range of approximately \$50 million–\$200 million. The \$10 million bid covers a combined package of data and software, and the public record does not allocate the price between them. Spirit’s Capability Alpha has not been measured, so these are conditional break-even and scale calculations, not an appraisal. The framework also considers buyer concentration, restrictive rights, and preemption as potential sources of divergence between observed price and buyer-specific value.

Keywords: data valuation; AI training data; model evaluation; intangible assets; reverse DCF; industrial organization.

1 Introduction

Proprietary enterprise records are becoming inputs to model training, post-training, and evaluation. These assets include expert demonstrations, operational workflows, internal software repositories, communications, transaction records, and linked records of decisions and outcomes. Enterprise information systems have produced such records for decades as event logs and audit trails that record how business processes unfold; Aalst [1] surveys this process-mining tradition. What is changing is their role in AI production. Modern agent training can use multi-step trajectories as demonstrations, preference data, evaluation episodes, or environment-grounded supervision, making the sequence and linkage between context, action, tool use, and outcome economically relevant in a way that a static document or feature table may not capture. Chen and Yuille [2] and Yang et al. [3] provide recent examples of agent post-training from expert or environment-grounded trajectories. Oderinwale and Kazlauskas [4] document the broader shift toward proprietary AI training data and identify context-dependent value and contribution to AI production as open measurement problems. Company-reported studies by Mercor [5] and Harvey [6] also show that expert data can

An active market is already visible. Mercor [7] and Scale AI [8] market enterprise workflow data for frontier-model development, while publicly reported licenses provide direct price signals: Reddit-Google, News Corp-OpenAI, New York Times-Amazon, and News Corp-Meta each involve reported consideration in the tens of millions of dollars per year or more [A1, A4b, A7, A10]. The transactions differ materially in rights and intended use, but together show that proprietary AI data and content access is already being priced at economically material levels. The value of any particular archive remains an empirical question.

The valuation problem draws on several established literatures. Coyle and Manley [9] and OECD [10] review economic approaches to data valuation, while the International Valuation Standards Council [11, 12] provides established machinery for valuing intangible assets. Howard [13] links information to the economic consequences of decisions. Dalessandro, Perlich, and Raeder [14] directly estimate the economic value of incremental predictive data by translating changes in model-performance metrics into expected economic gain. On the machine-learning side, Ghorbani and Zou [15], Jia et al. [16], Xu, Luo, and Li [17], and Tian et al. [18] provide contribution-based or utility-based approaches to data valuation. OpenAI [19, 20] reports evaluations of model performance on tasks with observable economic context.

The contribution of this paper is an AI-specific integration and transaction framework that combines controlled evaluation of proprietary enterprise corpora with buyer-specific intangible-asset valuation, economic life, rights, market structure, and an observed-price inversion. The performance-to-economic translation follows established prior work, including Dalessandro, Perlich, and Raeder [14]. Two quantities organize the analysis. *Capability Alpha* is the task-level performance gain attributable to a proprietary dataset under a specified evaluation design. *Price-Implied Uplift* reverses a revenue-based discounted-cash-flow model to recover the persistent uplift required to justify an observed data price. Empirical estimates of Capability Alpha should be accompanied by confidence intervals or other uncertainty measures appropriate to the evaluation design.

The framework is applied to Spirit Aviation Holdings’ proposed transfer of deidentified data and software assets to Google LLC. Spirit Aviation Holdings, Inc. [21] identifies Google as the successful bidder at \$10 million, subject to court approval. The public filing describes a large operational corpus spanning communications, IT records, source code, pricing, revenue, airline operations, maintenance, finance, legal, and employee records. The selected bid covers both data and software, so it is an observable package price. It does not identify a standalone price for data.

Two quantitative exercises are used. The first is a forward sensitivity analysis: given assumptions

about economic impact and value capture, what present value follows for a particular buyer? The second is PIU: given the selected bid and a stated revenue base, how much persistent revenue-equivalent uplift is required to break even? Under an illustrative 5% Google Cloud revenue-scope assumption, the \$10 million bid implies a PIU of about 0.24% over three years. A separate forward sensitivity calculation produces a \$50 million–\$200 million range under the assumptions stated in Section 6.2.

Market structure is treated separately from direct use value. Gans [22] and Korinek and Vipra [23] describe concentration and complementary inputs in AI markets, including compute, model development, and distribution. Restrictive rights can also create strategic value when rival access has a measurable competitive effect. These mechanisms can cause observed price to diverge from buyer-specific economic value.

2 Related literature and positioning

2.1 Intangible assets and data valuation

The valuation literature already provides the financial machinery required to value intangible assets. Cost approaches estimate the resources required to create or replace an asset. Market approaches use comparable transactions. Income approaches discount incremental cash flows attributable to an asset. The International Valuation Standards Council [11, 12] distinguishes Market Value from Investment Value or Worth, which is specific to an owner or prospective owner, and Synergistic Value, which can arise from combinations of assets or interests.

Coyle and Manley [9] find no single empirical valuation method that generalizes across data types and uses. OECD [10] likewise emphasizes that value depends on access, governance, rights, and context of use. Jones and Tonetti [24] formalize the nonrival nature of data and show why ownership, reuse, and market structure can affect whether data is shared or retained. Mohan, Bharathy, and Jalan [25] review enterprise-data valuation methods and document continuing implementation and reporting challenges. Bhutani, Ordoñez, and Veldkamp [26] model data as a productive asset whose value can depreciate, reinforcing the need to represent economic life explicitly.

The AI-training-data literature makes the production problem explicit. Oderinwale and Kazlauskas [4] characterize data as a distinct input to AI production and identify context-dependent value and estimation of data’s contribution to production as core open questions. Income-based valuation can translate an attributable economic benefit into value, but it requires a defensible estimate of the benefit first. Rappaport and Mauboussin [27] provide the reverse discounted-cash-flow logic used here to start from price and infer the operating performance required to justify it.

2.2 Economic value of information and incremental predictive data

The closest antecedent to the technical-to-economic bridge in this paper predates modern foundation models. Howard [13] assigns economic value to information through the decisions and consequences it changes. Dalessandro, Perlich, and Raeder [14] estimate the economic value of incremental commercial data by first measuring the change in predictive-model performance and then translating that change into expected economic gain. This is direct prior art for the performance-to-value step and establishes that data value is use-dependent.

The present framework applies that logic to proprietary enterprise corpora used in model adaptation, post-training, retrieval, evaluation, and AI systems; heterogeneous task families and model

stacks; legal and contractual usability; technological obsolescence; buyer-specific complements; and thin markets in which restrictive rights may create strategic effects. Bostoen and Monti [28] similarly emphasize transaction design and bargaining institutions when direct comparables for AI training data are scarce. Capability Alpha is therefore a named empirical input to the framework. Causal inference and the general performance-to-economic translation step are established prior art; the contribution here is the transaction-level integration described above.

2.3 From predictive features to workflow-linked trajectories

Enterprise workflow data has a long history as an internal analytical asset. Aalst [1] describes event logs as records of business-process execution used for process discovery, conformance checking, and operational analysis. The recent change is in the production role assigned to those records. In agentic systems, a linked workflow can become training supervision or an evaluation environment when it preserves enough of the sequence from state and context through action, tool use, intermediate results, and outcome.

Recent technical work makes this distinction concrete. Chen and Yuille [2] treat expert trajectories as supervision for agent policy optimization and convert multi-turn demonstrations into state-conditioned preferences. Yang et al. [3] show that interaction trajectories exposing inspect-act-verify structure can produce strong generalization and data efficiency in terminal-agent post-training. These results do not establish the monetary value of enterprise records. They show why linkage and trajectory structure can matter technically for modern agent training.

Operational records themselves are longstanding. The shift lies in the data object that is being priced and in its role inside the AI production process. Dalessandro, Perlich, and Raeder [14] study incremental third-party feature data for defined predictive applications in advertising and direct mail. Modern enterprise AI can assign value to workflow-linked records that were originally generated as operational exhaust, process evidence, communications, or audit trails, because the same corpus may support demonstrations, evaluation tasks, verifiers, retrieval, and post-training across multiple task families. Current market offerings from Mercor [7] and Scale AI [8] provide direct evidence that such workflow data is now being packaged for external AI buyers. The valuation problem addressed here is the external pricing of that emerging AI use while preserving the established insight that data value is conditional on the buyer, task, and decision environment.

2.4 Machine-learning data valuation

Machine-learning research approaches data value from the technical side. Ghorbani and Zou [15] assign value to training observations according to their marginal contribution to a model-performance payoff, and Jia et al. [16] develop more efficient Shapley-based estimators. Related work studies the effect of changing the training set itself. Ilyas et al. [29] fit *datamodels* that predict how model outputs change across training-set subsets, while Park et al. [30] develop TRAK for scalable attribution of model behavior to training examples. In the LLM setting, Xia et al. [31] use influence estimates to select instruction-tuning data targeted at a desired capability. These methods are useful for locating influential subsets and approximating data counterfactuals.

They do not remove the need for a direct asset-level experiment. Capability Alpha concerns the effect of giving a specified development or deployment procedure access to dataset D as a treatment. Training-data attribution can help decide which parts of D to test, diagnose where gains come from, or reduce the cost of subset search; the final estimate used for valuation should come from a matched treatment-control evaluation whenever that experiment is feasible.

Recent work also addresses buyer-specific utility and information asymmetry in AI data transactions. Yang, Gao, and Zheng [32] propose a privacy-preserving protocol that scores external data against a buyer’s model without revealing the raw data, directly addressing the inspection problem that arises when utility must be assessed before exchange. Xu, Luo, and Li [17] combine information-density measures, proxy-model training gain, influence methods, and Data Shapley for utility-aware pricing of LLM training data. These approaches strengthen the technical measurement layer. A contribution measured in benchmark or influence units still requires a separate mapping into buyer-specific economics.

Capability Alpha is used here as that intermediate empirical quantity. It is task- and evaluation-specific, and the valuation layer translates it into buyer-specific economic outcomes; benchmark points have no universal dollar price.

2.5 Evaluation on economically relevant tasks and expert data

Economically grounded evaluations make the measurement step more plausible. OpenAI [19] reports more than 1,400 SWE-Lancer freelance software-engineering tasks associated with roughly \$1 million in actual payouts. OpenAI [20] evaluates 1,320 GDPval tasks across 44 occupations selected for economic significance using work products created by experienced professionals. These benchmarks move evaluation closer to tasks whose economic stakes can be observed or approximated.

Company-reported post-training experiments provide examples of the causal-attribution problem. Mercor [5] reports large gains after training GLM 4.6 on 874 expert-created tasks in professional environments. Harvey [6] reports gains from a stack combining public, synthetic, and human-expert data with post-training and harness changes, and separately shows that harness changes alone can move benchmark performance. These reports are useful as examples of measurement design; they are not independent estimates of the value of enterprise data.

A before-and-after comparison is insufficient when data, model checkpoint, tools, compute, rewards, and harness design move together. The empirical object needed for valuation is the marginal contribution of the proprietary corpus under a controlled stack.

2.6 Economics of AI and buyer concentration

The same measured capability gain can have different economic value across buyers. Demirer et al. [33] document substantial variation in demand for model capability across applications using OpenRouter and Microsoft Azure usage data. Gans [22] emphasizes that data markets interact with bottlenecks such as compute, model openness, and platform design in determining AI market power. Korinek and Vipra [23] analyzes scale economies and concentration in foundation-model development, including the role of compute, data, and talent.

For data transactions, this means the set of firms able to receive a dataset can be much larger than the set able to monetize it at scale. Buyer-specific values can differ substantially, and thin buyer markets can make observed prices sensitive to bargaining power, process design, and the value of the next credible bidder.

2.7 Exclusivity, foreclosure, and preemption

Exclusive or restrictive rights add a strategic channel. OECD [34] treats access to data and input foreclosure as potentially relevant competitive mechanisms in digital mergers. OECD [35] similarly

discusses data access, vertical integration, model restrictiveness, and control of enabling inputs in AI downstream competition.

Preemption also appears in models of strategic investment. Weeds [36] analyzes R&D investment under uncertainty when competition for a winner-takes-all asset changes investment timing through preemption incentives. In a data transaction, the narrower analogue is that restrictive rights can create value for a buyer if denying a close rival access to the same capability-relevant input changes expected profits.

The empirical framework separates direct use value from any strategic component. The Spirit calculations below include direct use value only because the public record does not provide enough evidence to estimate a competitive-denial effect.

2.8 Research gap

The component literatures already cover much of the underlying machinery. The gap addressed here is narrower. Earlier work establishes the economic value of information and the translation of incremental predictive performance into money; process mining establishes the analytical value of enterprise event logs; recent AI-data work develops contribution scoring and privacy-preserving buyer-side utility measurement; and agent-training research establishes the technical value of interaction trajectories. What remains underdeveloped is a transaction-level framework for workflow-linked enterprise corpora used in modern AI production: a controlled estimate of incremental capability across economically relevant tasks, an explicit mapping into buyer-specific economics, treatment of useful life and rights, and an inversion of an observed price into a testable break-even threshold. The Spirit case is used to show how these pieces can be stated together without treating any sensitivity scenario as an appraisal.

3 Framework

3.1 Measurement concepts

Three concepts organize the measurement layer.

Capability gap. For task family t , let H_t denote a reference level of performance and F_t the performance of a defined practical baseline under a standardized harness. The gap $G_t = H_t - F_t$ is a diagnostic of how much task-level performance remains available to improve. A large gap creates room for improvement; it says nothing by itself about whether dataset D contains useful signal.

Capability Alpha. Capability Alpha is a task-level treatment effect measured under a declared reference adaptation stack. For dataset D , task t , and a pre-specified data dose k , let M_0 and M_1 denote the matched control and treatment models defined in Section 4. We define

$$CA_{D,t}(k) = \mathbb{E}[Y_t(M_1)] - \mathbb{E}[Y_t(M_0)], \quad (1)$$

where Y_t is a pre-specified held-out task metric and the expectation is over training and evaluation randomness. The reference model, adaptation method, token and compute budget, and evaluation harness are fixed before the comparison. Capability Alpha is therefore conditional on a disclosed experimental stack, in the same sense that contribution-based data valuation depends on a learning algorithm and utility metric [15, 16]. It can be positive or negative. Different task metrics are kept separate and translated into economic units task by task through $\lambda_{b,t}$; no cross-task technical average is used.

Verifiability and outcome linkage. These are descriptive corpus properties and are not separate valuation variables. Outcome linkage is related to the event-log and case-linkage concepts described by Aalst [1], where enterprise records are connected into executions of a business process. Here the emphasis is on whether the recovered sequence is usable for AI evaluation or post-training. Verifiability concerns the share and quality of cases for which success can be judged by an objective, expert, or reproducible criterion. Outcome linkage concerns the ability to connect context, action, intermediate state, and eventual outcome across records. These properties can matter because linked trajectories can supply demonstrations, environments, preference signals, and reward or verifier targets, as illustrated by Chen and Yuille [2] and Yang et al. [3].

Figure 1 summarizes the measurement-to-valuation pipeline.

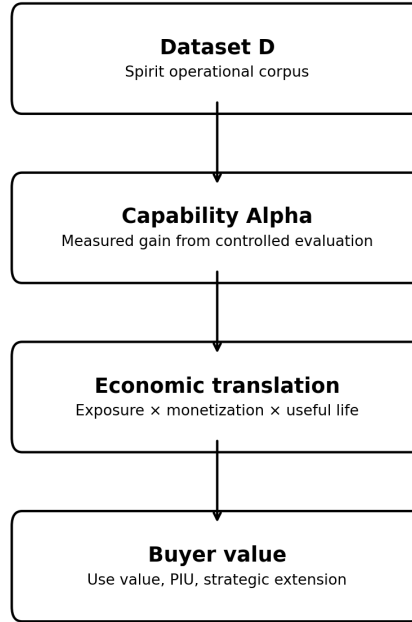


Figure 1: From proprietary data to buyer-specific value.

3.2 Direct use value

For buyer b , task family t , and future year h , let $E_{b,t,h}$ denote annual economic exposure measured in dollars of revenue, cost, or another defined economic base. Let $\lambda_{b,t}$ map one unit of the task-performance metric into a fractional change in that economic base. Let $d_{D,t,h}$ denote the fraction of the initial data advantage expected to persist at horizon h . A reduced-form direct use value is

$$V_b^{use}(D) = \rho_D \sum_{h=1}^T \frac{1}{(1+r)^h} \sum_t (E_{b,t,h} \lambda_{b,t} C A_{D,t} d_{D,t,h}) - C_b(D), \quad (2)$$

where $\rho_D \in [0, 1]$ is a reduced-form rights-and-usability adjustment, r is the annual discount rate, and $C_b(D)$ is the buyer-specific cost of acquisition, legal work, deidentification, cleaning, training, integration, and deployment. The factor ρ_D should represent restrictions on usable rights and

access; technical uncertainty belongs in the Capability Alpha estimate, and strategic exclusivity is handled separately below. When a legal restriction removes an entire use case, the corresponding task exposure should be set to zero instead of relying on a single global haircut. Under these definitions, the inner product has units of dollars per year. The $\lambda_{b,t}CA_{D,t}$ term is a local linear approximation. If commercial response is thresholded or strongly nonlinear, it should be replaced by an estimated response function. Task-level economic bases must also be defined to avoid double counting overlapping revenue or cost exposure.

The distinction between r and $d_{D,t,h}$ is important. The discount rate converts future cash flows into present value and reflects time value, risk, and opportunity cost. The persistence term describes whether the data advantage itself erodes as models improve, substitute data appears, or business processes change, consistent with economic-life treatment in the International Valuation Standards Council [12] and data-depreciation analysis by Bhutani, Ordoñez, and Veldkamp [26].

The same equation can be applied to the current owner by setting b equal to the owner. Capability Alpha remains the measured technical contribution under the reference evaluation, while owner or buyer heterogeneity enters through economic exposure, monetization, costs, and the rights package. This allows internal use and transfer to be compared within the same valuation structure.

3.3 Price-Implied Uplift

For the Spirit case, we also use a simplified revenue-base specialization. Let R_{scope} be a defined annual revenue base, m the operating margin applied to incremental revenue, T the useful life, r the discount rate, ρ the rights adjustment, and C_b buyer-specific integration and transaction costs. Define the annuity factor

$$A(T, r) = \sum_{h=1}^T \frac{1}{(1+r)^h}.$$

The revenue-equivalent uplift u required to support transaction price P is

$$\text{PIU} = u^* = \frac{P + C_b}{R_{\text{scope}}mA(T, r)\rho}. \quad (3)$$

This is a reverse discounted-cash-flow calculation applied to a data transaction, following the price-implied expectations logic of Rappaport and Mauboussin [27]. In this specialization, PIU is a percentage uplift to the scoped revenue base, with operating margin converting incremental revenue into operating profit. A cost-savings application would use a corresponding cost base and conversion rule instead of mechanically applying an operating margin.

PIU reports the break-even threshold implied by the stated revenue-base scenario. Appendix A applies the same inversion to four additional public AI-data/content transactions, testing its portability across observed prices and contract structures. Realized Capability Alpha remains a separate empirical quantity.

3.4 Strategic extension

Direct use value can be extended when a separately identifiable strategic effect is supported by evidence:

$$V_b^{\text{total}}(D) = V_b^{\text{use}}(D) + S_b(D). \quad (4)$$

Here $S_b(D)$ is the present value, to buyer b , of named strategic effects that are outside the buyer's own task-level cash flows in Equation 2. Each component included in $S_b(D)$ must correspond to a stated strategic mechanism and be estimated separately.

One possible component is preemption value. For buyer b ,

$$V_b^{\text{preemption}}(D) = \Pi_b(\text{rivals without access to } D) - \Pi_b(\text{rivals with access to } D). \quad (5)$$

A separately estimated option value can also enter $S_b(D)$ when ownership creates a credible future use that is not represented in the enumerated task cash flows. The Spirit sensitivity range sets $S_b(D) = 0$ because the public record does not support an estimate of either component.

4 Estimating Capability Alpha

The empirical object is a dataset-level counterfactual: under a fixed model-adaptation procedure and fixed resource budget, how much does replacing available reference data with proprietary data D change out-of-sample performance on task t ? This follows the use-dependent utility logic of Data Shapley [15, 16], the training-set counterfactual view of Datamodels [29], and recent LLM work that prices data from empirical training gain [17]. Capability Alpha is estimated by direct matched retraining; influence and attribution methods are used only as secondary diagnostics.

4.1 Estimand and reference design

The measurement fixes a base model M , an adaptation operator $\mathcal{A}$, a total adaptation budget B , a reference corpus R , a proprietary-data dose $D_k \subseteq D$ of size k , and a held-out task evaluation $\mathcal{E}_t$ with metric Y_t . The reference corpus represents the data that would occupy the same adaptation budget if the proprietary asset were unavailable.

The adaptation operator is the mechanism through which training data can change model capability. It may be full fine-tuning or a parameter-efficient method such as LoRA [37]; whichever procedure is chosen is held fixed across treatment and control. The resulting Capability Alpha is therefore conditional on a disclosed reference stack; an unspecified notion of “data quality” is insufficient.

For training seed s , define

$$M_0^{(s)} = \mathcal{A}(M, R_B; s), \quad (6)$$

$$M_1^{(s)} = \mathcal{A}(M, R_{B-k} \cup D_k; s), \quad (7)$$

where R_B contains B units of reference data and R_{B-k} contains $B - k$ units drawn under the same sampling rule. Both arms use the same checkpoint, optimizer, adaptation method, training schedule, token or example budget, compute budget, and evaluation harness. The intervention changes only the provenance of k units of training data. Comparing an unadapted model with a model trained on D would mix the effect of the data with the effect of additional optimization.

4.2 Held-out task construction

The task family t , target distribution, metric Y_t , and verifier are fixed before treatment results are observed. Economically relevant tasks are preferred because the subsequent valuation requires a mapping from technical performance into business outcomes. SWE-Lancer and GDPval provide examples of evaluations built from realistic work with explicit success criteria [19, 20].

The evaluation set $\mathcal{E}_t$ is disjoint from the adaptation data. For workflow-linked enterprise corpora, splitting occurs at the case or entity level needed to prevent leakage: an incident, customer,

project, route, contract, or later time period is held out as a unit. Objective outcomes are used as verifiers when available; otherwise a calibrated expert rubric is fixed in advance. Near duplicates, copied resolutions, shared identifiers, and temporal overlap are screened because they can turn memorization into apparent capability. Datamodels shows how training-set counterfactuals can expose this form of leakage [29].

4.3 Treatment-effect estimator

Treatment and control are evaluated on the same held-out cases. With S matched training seeds,

$$\widehat{CA}_{D,t}(k) = \frac{1}{S} \sum_{s=1}^S \left[Y_t(M_1^{(s)}, \mathcal{E}_t) - Y_t(M_0^{(s)}, \mathcal{E}_t) \right]. \quad (8)$$

A confidence interval is estimated from the paired evaluation outcomes and run-to-run variation across seeds. Negative estimates are retained. A placebo arm using an equally sized irrelevant or shuffled corpus can be added when feasible to detect procedural gains unrelated to information in D .

As a stylized example, let D contain IT-support trajectories and let Y_t be verified resolution rate on 1,000 held-out incidents. Suppose both arms use the same LoRA configuration and a 20-million-token adaptation budget. The control uses 20 million reference tokens; the treatment replaces 5 million of those tokens with D . If mean resolution is 55.0% for control and 61.2% for treatment across matched runs, then $\widehat{CA}_{D,t}(5M) = +6.2$ percentage points. That number is the measured technical contribution of the proprietary data under the declared stack. It contains no monetary assumption.

4.4 Scaling, transfer, and valuation handoff

Repeating the experiment at pre-specified values of k produces a capability curve $CA_{D,t}(k)$ and reveals saturation. Repeating the matched intervention on a second model family or checkpoint tests whether the sign and approximate magnitude transfer beyond one reference model. Regressions on adjacent economically relevant tasks are measured alongside the target-task gain.

Data Shapley [15], Datamodels [29], TRAK [30], and LESS [31] can then rank records or trajectories for follow-up ablations and data selection. They do not replace the direct treatment-control estimate in Equation 8. For a corpus spanning several task families, the measurement output is the set of task-specific estimates and uncertainty intervals $\{\widehat{CA}_{D,t}(k), CI_{D,t}(k)\}_{t \in \mathcal{T}}$. Each task is carried separately into Equation 2, where $E_{b,t,h}$ and $\lambda_{b,t}$ translate the measured capability contribution into buyer-specific economic value.

5 Spirit Aviation case study

5.1 Transaction and asset

Spirit Aviation Holdings, Inc. [21] designated Google LLC as the successful bidder for a set of data and software assets at \$10 million, subject to court approval and the required deidentification process. The same filing identifies Mercor as the \$7.5 million alternate bidder. Business Insider [38] later reported micro1’s \$12.5 million offer outside the initial auction process. As of August 23, 2026, approval remained pending after employee-privacy objections delayed the sale hearing,

as reported by Bloomberg Law [39]. Bloomberg Law [40] reported Google’s statement that the acquired portion of Spirit’s enterprise dataset could help improve its products and AI models.

Spirit Aviation Holdings, Inc. [21] describes the selected \$10 million bid as covering a combined package of data and internally developed software. The public record does not allocate the package price between those components. The PIU exercise below uses the full \$10 million as P . If the software has positive standalone value, that convention places an upper bound on the price attributed to the data component; if the components are complementary, a clean allocation may not be economically meaningful. Spirit Aviation Holdings, Inc. [21] also requires Google to bear third-party deidentification costs in addition to the headline bid, so the zero-cost PIU case is explicitly a base sensitivity and does not represent total acquisition cost.

The asset schedule in Spirit Aviation Holdings, Inc. [21] describes a broad operating corpus. It includes approximately 100 million emails, 500 million Microsoft Teams items, 17.1 million OneDrive items, 20.6 million SharePoint items, 667,563 IT tickets, 516 code repositories containing roughly 30 million lines of code, 372,585 commits, 3.53 billion booking-curve observations, 7.25 billion competitor-flight observations, 7.51 billion revenue transactions, and extensive airline operations, maintenance, finance, legal, and employee records.

The agreement in Spirit Aviation Holdings, Inc. [21] requires deidentification while preserving referential integrity across the data structure. Stable identifiers and timestamps could preserve useful workflow trajectories, for example linking a pricing state to a booking outcome or an IT ticket to a code change and deployment result. Whether those linkages are complete enough to support training or evaluation is an empirical question that cannot be answered from the asset schedule alone.

The transferred assets are a bounded subset of Spirit’s data estate. Spirit Aviation Holdings, Inc. [21] excludes major consumer datasets, including customer profiles and Free Spirit member records.

5.2 Why the asset could matter to Google

Google LLC is the buyer. Google Cloud is used below as one observable monetization base because Alphabet Inc. [41] publicly reports its revenue and operating margin. Cloud serves solely as a transparent economic base for the sensitivity analysis.

Google Cloud already supports airline optimization and decision-support use cases. Google Cloud [42] describes operational decision support using crew, passenger, rotation, and technical data and reports savings from rotation optimization. Google Cloud [43] describes machine-learning revenue management and reports material revenue effects in customer deployments. These examples establish that airline operations and revenue-management tasks can have measurable commercial consequences. They provide no estimate of Spirit’s contribution.

Spirit valuation scenarios

6.1 Observed bids

The observed process gives three price signals. Spirit Aviation Holdings, Inc. [21] records Mercor at \$7.5 million and Google selected at \$10 million; Business Insider [38] reports micro1’s later \$12.5 million indication. The micro1 figure is stated willingness to pay because it arrived outside the initial auction process and had not become a completed transaction in the record used here.

Figure 2 places these market observations next to the forward sensitivity range developed below. The observed bids and the modeled interval answer different questions.

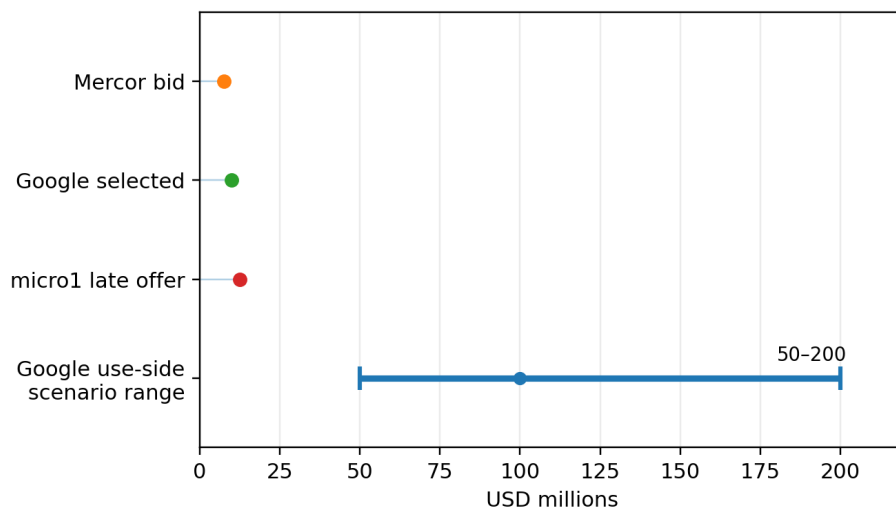


Figure 2: Observed bids and one conditional buyer-specific use-value range.

6.2 Illustrative forward sensitivity, not an appraisal

The \$50 million–\$200 million range is generated by the assumptions in Table 1. Observed inputs and scenario inputs are shown separately.

Input	Value	Status
2026 global airline revenue	\$1.165T	IATA [44]
Modeled improvement in airline economics	1.0%	Scenario assumption
Google capture of created value as incremental revenue	0.5%–2.0%	Scenario assumption
Google Cloud operating margin	35.6%	Alphabet [41]
Economic life	3 years	Scenario assumption
Discount rate	12%	Scenario assumption; not a Google WACC estimate

Table 1: Inputs to the forward Spirit use-value sensitivity calculation.

A 1% improvement in airline economics corresponds to \$11.65 billion of annual value across the industry. If Google captures 0.5%–2.0% of that value as incremental revenue, the implied annual Google revenue is approximately \$58.25 million–\$233 million. Applying the 35.6% Google Cloud operating margin gives approximately \$20.7 million–\$82.9 million of annual operating profit. A three-year life discounted at 12% has an annuity factor of approximately 2.402, producing a present value of approximately \$49.8 million–\$199.2 million.

The calculation is a scale sensitivity. The 1% industry effect and 0.5%–2.0% capture rates are assumptions. They are not estimated from Spirit data and should not be interpreted as a central case. Table 2 shows how strongly the result moves when the airline-impact assumption changes.

Modeled airline economic improvement	0.5% capture	1.0% capture	2.0% capture
0.25%	\$12.5M	\$24.9M	\$49.8M
0.50%	\$24.9M	\$49.8M	\$99.6M
1.00%	\$49.8M	\$99.6M	\$199.2M
2.00%	\$99.6M	\$199.2M	\$398.4M

Table 2: Present-value sensitivity to the two unobserved forward assumptions. Values use the observed Google Cloud margin, a three-year life, and a 12% discount rate.

The \$50 million–\$200 million interval is therefore one row of a wider sensitivity surface, not a valuation estimate. A measured Capability Alpha and a production estimate of λ would be required to replace these assumptions with evidence.

6.3 Price-Implied Uplift for Spirit

PIU starts from the selected \$10 million package bid. Alphabet Inc. [41] reports Q2 2026 Google Cloud revenue that annualizes to approximately \$99.072 billion. Define a *relevant revenue scope* as the share of annual Google Cloud revenue associated with products or workflows that could plausibly experience an economic effect from capabilities affected by the dataset.

For the illustrative 5% scenario,

$$R_{\text{scope}} = 0.05 \times \$99.072\text{B} = \$4.954\text{B}.$$

The 5% figure refers only to modeled scope; it carries no claim about aviation’s share of Cloud revenue or a 5% Spirit-driven revenue increase.

Applying the reported 35.6% Google Cloud segment operating margin as a proxy gives approximately \$1.763 billion of annual operating profit associated with the scoped revenue base. The segment margin is not an observed incremental margin for a Spirit-derived product, so it is another sensitivity input. Discounting three years at 12% gives a present-value profit base of about \$4.23 billion. With zero integration cost and $\rho = 1$ for this base sensitivity,

$$\text{PIU} = \frac{\$10\text{M}}{\$4.23\text{B}} \approx 0.236\%.$$

Table 3 and Figure 3 show the sensitivity to the scope assumption.

Relevant Cloud revenue scope	Annual revenue base	PIU for \$10M price
1%	\$0.991B	1.181%
2.5%	\$2.477B	0.472%
5%	\$4.954B	0.236%
10%	\$9.907B	0.118%

Table 3: Price-Implied Uplift under alternative revenue-scope assumptions.

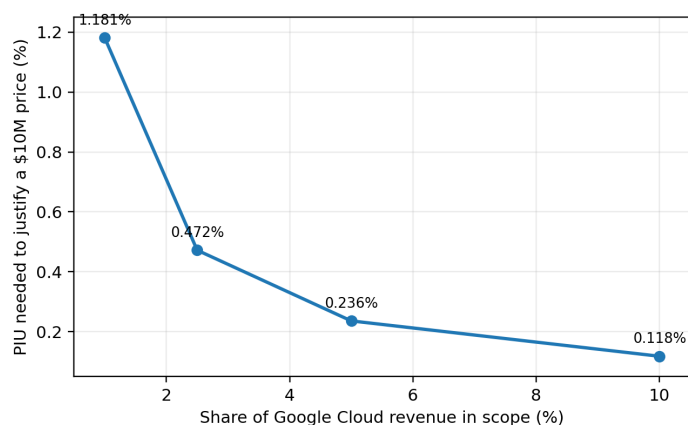


Figure 3: PIU declines as the modeled revenue scope grows.

Under the 5% scope assumption, a persistent 0.236% revenue-equivalent uplift in the scoped Google Cloud base is sufficient for a \$10 million purchase to break even over three years under the stated margin, life, rights, and cost assumptions. Holding the 5% scope and 35.6% margin fixed, changing the discount rate from 8% to 20% moves PIU only from about 0.220% to 0.269%; the revenue-scope and economic-translation assumptions are materially more consequential. Whether the dataset can generate the required effect is the empirical question.

The additional transactions sharpen the interpretation of Spirit's PIU. Applying the same reverse calculation to three recurring AI-data/content licenses involving mature public-company buyers produces the following standardized thresholds at a 5% economic scope. The detailed transaction terms, derivations, and full scope sensitivities are reported in Appendix A.

Transaction	Reported consideration	5% scope PIU
Spirit - Google	\$10M one-time package	0.236%*
New York Times - Amazon	\$20M-\$25M / year	0.58%-0.73%
News Corp - Meta	up to \$50M / year	<= 1.20%
Reddit - Google	~\$60M / year	1.42%

* Spirit uses the case-specific Google Cloud denominator and three-year DCF; the recurring-license rows use a standardized buyer operating-income proxy.

The contracts are not direct valuation comparables: they differ in rights, duration, refresh, intended use, economic base, and asset composition. The panel nevertheless provides a first empirical calibration of PIU. Across the three recurring public-company cases, the standardized 5% threshold is approximately 0.58%-1.42%. Spirit's case-specific 0.236% threshold falls below that range. This does not establish that Spirit was underpriced, but it means the selected price sets a comparatively low economic hurdle under the stated assumptions.

The exercise also shows that the economically exposed business surface matters more than a mechanical buyer-wide denominator. For the New York Times-Amazon license, using 5% of AWS rather than consolidated Amazon operating income raises the midpoint PIU from about 0.66% to 1.13% (Appendix A.4).

7 Market structure, retention, and strategic value

The Spirit process illustrates why transaction price and buyer-specific value can diverge. A proprietary corpus can be technically transferable to many firms while only a small number possess the models, compute, post-training infrastructure, distribution, and commercial exposure needed to monetize it at scale, consistent with the market-structure analyses of Gans [22] and Korinek and Vipra [23]. In a thin buyer market, a high-value buyer can obtain substantial surplus even when the observed price is much lower than its own Investment Value or Worth.

The current owner can be evaluated with the same framework. Setting b equal to the owner produces an internal retention value. A firm with limited economic exposure, weak commercialization channels, or high integration costs may rationally assign low internal value to a dataset even when the dataset has measurable Capability Alpha. A buyer with larger complementary assets can value the same measured gain much more highly. This difference creates a potential gain from trade without requiring a buyer-specific definition of Capability Alpha.

Transfer can also impose a competitive cost on the seller if the buyer, or customers of the buyer, use the resulting capability against the seller. That cost affects the seller’s reservation value and the contract terms it should accept. Field-of-use restrictions, nonexclusive licensing, time limits, or higher consideration can change that trade-off. The relevant strategic effect belongs in the rights package or in a separately estimated strategic component, not inside Capability Alpha.

Exclusivity can create preemption value when a close rival would gain meaningful capability from the same data and cannot obtain an equivalent substitute. Related mechanisms appear in OECD [34], OECD [35], and Weeds [36]. Equation 5 provides a way to state that effect. The Spirit sensitivity range excludes preemption because the public record does not provide enough evidence to estimate it.

Technology can change both sides of the market over time. Improvements in models, hardware, and post-training tooling can lower the complementary investment required to exploit a corpus and expand the set of economically capable buyers. The same progress can reduce the use value of a fixed corpus if baseline performance closes the relevant task gap or substitute data and synthetic environments improve. Asset value and seller bargaining power can therefore move in different directions.

8 Limitations and falsification

Spirit’s Capability Alpha has not been measured. The available evidence therefore cannot establish whether the assets were underpriced. The framework identifies the technical and economic quantities required to evaluate that claim.

Capability Alpha is conditional on the evaluation stack. Model family, compute, prompting, tools, retrieval, reward design, and harness engineering can all affect the observed treatment effect. Robust estimation requires controlled ablations, uncertainty intervals, and transfer tests across model families. Company-reported post-training results cited in Section 2 illustrate these design issues but are not substitutes for an independent Spirit evaluation.

Economic translation is a second major uncertainty. The coefficients $E_{b,t,h}$ and $\lambda_{b,t}$ determine how a technical gain becomes money, and neither is observed for Spirit. Equation 2 also uses a local linear and additive approximation; nonlinear product response, interactions across tasks, and overlapping economic bases require a richer production model. The forward \$50 million–\$200 million range is especially sensitive to the assumed 1% airline-industry effect and 0.5%–2.0% Google capture rate. The PIU calculation is algebraically tighter because it starts from an observed price, but its denominator still depends on the chosen revenue scope, margin, life, rights, and costs.

Rights and usability add further uncertainty. A technically useful corpus can lose economic value through privacy restrictions, deidentification burdens, broken linkage, contractual limits, or high integration costs. Economic life is uncertain because frontier progress and substitute data can erode scarcity, consistent with the International Valuation Standards Council [12] and Bhutani, Ordoñez, and Veldkamp [26].

The Spirit buyer-specific value thesis would weaken if representative samples show low practical usability, weak Outcome Linkage, low Verifier Density, little transfer to economically relevant tasks, or a measured Capability Alpha that translates into commercial uplift below the PIU threshold under defensible scope assumptions. The Spirit package also combines data and software, and Google bears deidentification costs beyond the headline bid. A standalone data valuation would therefore require an allocation of package value and a measured cost term. PIU remains usable as a package-level sensitivity. Those issues limit any claim about a standalone Spirit data price.

9 Conclusion

This paper adapts and combines established ideas from value-of-information analysis, predictive-data valuation, intangible-asset finance, and machine-learning data valuation for proprietary enterprise corpora used in contemporary AI systems. Capability Alpha is the paper’s shorthand for the controlled task-level performance contribution used as the technical input. Direct use value then depends on economic exposure, the mapping from performance to commercial outcomes, persistence, usable rights, and buyer-specific costs. PIU provides the inverse calculation for a revenue-base scenario by recovering the persistent revenue-equivalent uplift required to justify an observed transaction price.

Under the illustrative 5% Google Cloud scope, the selected \$10 million Spirit package implies a three-year PIU of about 0.24%; the forward sensitivity spans roughly \$50 million–\$200 million. Across three additional recurring transactions with mature public-company buyers, standardized 5% PIUs are approximately 0.58%–1.42%. Spirit’s case-specific threshold therefore falls below that range. The contracts are not direct comparables, so this does not establish underpricing, but it shows that the selected Spirit price sets a comparatively low economic hurdle under the stated assumptions. Appendix A reports the transaction-level derivations and sensitivities.

The same framework also clarifies the owner’s decision to retain, use, license, or sell data. The current owner and an external buyer can assign different values to the same measured Capability Alpha because their economic exposure, commercialization channels, costs, and strategic incentives differ. An important emerging category is workflow-linked operational data: records that predate modern AI as business-process exhaust, communications, audit trails, or process evidence, but can now be reused as demonstrations, evaluation episodes, verifiers, or post-training trajectories for agentic systems. Empirical progress depends on controlled data evaluations, credible production mappings, transparent rights packages, and repeated observations of actual transactions and realized outcomes.

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Appendix A. Public transaction calculations and source notes

A.1 Purpose, evidence hierarchy, and recurring-license derivation

This appendix applies PIU to observed or reported AI-data/content transactions beyond Spirit. It is designed to separate three evidentiary layers that are often conflated in discussion of the market: (i) terms disclosed by a contracting party or in a securities filing; (ii) the existence and stated purpose of a partnership disclosed by the parties; and (iii) consideration reported by a reputable financial-news source when the parties keep price confidential. A press-reported price is therefore labeled *reported* throughout. It is not treated as a contractual disclosure.

For a one-time purchase price P , the paper defines

$$\text{PIU} = \frac{P + C}{R_{\text{scope}} m A(T, r) \rho}, \quad A(T, r) = \sum_{h=1}^T \frac{1}{(1 + r)^h}.$$

For a recurring license with level annual consideration F and level attributable annual benefit over the same horizon, the present value of consideration is $FA(T, r)$. Substituting this payment stream gives

$$\text{PIU}_{\text{annual}} = \frac{FA(T, r) + C}{R_{\text{scope}} m A(T, r) \rho}.$$

When the base comparison sets $C = 0$, $\rho = 1$, and the fee and economic benefit have the same annual timing, the annuity factor cancels:

$$\boxed{\text{PIU}_{\text{annual}} = \frac{F}{R_{\text{scope}} m}} = \frac{F}{s R m},$$

where $R_{\text{scope}} = sR$. If the consolidated operating margin $m = \text{OI}/R$ is used only as a standardized proxy, then

$$\boxed{\text{PIU}_{\text{annual}} = \frac{F}{s \text{OI}}}.$$

This identity is convenient but does not make consolidated operating margin the economically correct margin for every AI use case. It provides a transparent cross-case baseline. The case sheets below use the most recent completed fiscal year available before each transaction. Scope sensitivity is reported at 1%, 2.5%, 5%, and 10% where a public-company buyer permits the calculation.

The broader literature supports treating these cases as heterogeneous rather than perfectly comparable. Oderinwale and Kazlauskas [4] document AI training-data deals from 2020–2025 and identify multiple pricing mechanisms. Bostoen and Monti [28] emphasize that bilateral licensing terms are sparse and institution-dependent. A separate review of 83 known commercial content agreements likewise finds that public terms are frequently incomplete [A12]. The purpose of the exercise is therefore not to infer a universal “price per token” or “price per article.” It is to test whether observed consideration can be converted into an explicit buyer-side break-even hypothesis.

A.2 Reddit → Google, 2024

Transaction evidence. Reuters reported on February 22, 2024 that Reddit had licensed its content to Google for approximately \$60 million annually and that Google could use Reddit content to train AI models [A1]. Reddit’s S-1, filed the same day, independently disclosed that in January

2024 it had entered data-licensing arrangements with aggregate contract value of \$203.0 million, terms of two to three years, and at least \$66.4 million expected to be recognized in 2024 [A2]. The filing also stated that substantially all contract value associated with licensing revenue was derived from one partner and described model training as a licensing use. The S-1 does not name Google or state a Google-specific price, so the \$60 million annual figure remains a Reuters-reported term rather than an issuer disclosure.

Buyer baseline. Alphabet’s FY2023 Form 10-K reports \$307.394 billion of revenue and \$84.293 billion of operating income, a consolidated operating margin of approximately 27.42% [A3]. With annual fee $F = \$60$ million and standardized exposure $s = 5\%$,

$$\text{PIU}_{5\%} = \frac{\$60\text{M}}{0.05 \times \$84.293\text{B}} = 1.424\%.$$

Standardized Alphabet scope	Scoped FY2023 operating income	PIU at \$60M/year
1%	\$0.843B	7.118%
2.5%	\$2.107B	2.847%
5%	\$4.215B	1.424%
10%	\$8.429B	0.712%

Table A1. Reddit–Google recurring-license PIU sensitivity using Alphabet FY2023 consolidated operating income as a standardized proxy.

The calculation says only that a \$60 million annual fee requires approximately 1.42% of persistent revenue-equivalent improvement if 5% of Alphabet’s pre-deal operating-income base is the relevant exposure. A narrower true economic base raises the threshold; a broader base lowers it. The licensed object also differs from Spirit: Reddit provides a continuously renewing platform corpus rather than a bounded enterprise archive.

A.3 News Corp → OpenAI, 2024

Transaction evidence. News Corp and OpenAI announced a multi-year global partnership on May 22, 2024 covering current and archived content from major News Corp publications, with permission to display content in responses and enhance OpenAI products [A4]. Reuters reported that the agreement could be worth more than \$250 million over five years [A5]. The Wall Street Journal reported that the consideration includes both cash and credits for OpenAI technology, and that the content may be used both to answer queries and train OpenAI technology [A4b]. The reported contract value is therefore not a pure cash data price.

Why no point PIU is reported. OpenAI was private and did not publish audited product-level economics. Later 2024 reporting described approximately \$3.7 billion of expected 2024 revenue and an approximately \$5 billion loss [A5b]. A current consolidated operating-margin denominator would consequently be negative and is not a meaningful measure of the economic base for a rapidly growing AI product. This is a case where forcing Equation 3 into a point estimate would create false precision.

A transparent sensitivity can still be stated. Treating \$250 million over five years as a nominal annualized contract-value floor of \$50 million per year, ignoring the cash/credit mix and growth for illustration, gives

$$\text{PIU}^{\text{illustrative}} = \frac{\$50\text{M}}{s \times \$3.7\text{B} \times m^+},$$

where m^+ is an assumed positive contribution margin rather than OpenAI’s reported consolidated operating margin.

Revenue scope s	$m^+ = 25\%$	$m^+ = 50\%$	$m^+ = 75\%$
25%	21.62%	10.81%	7.21%
50%	10.81%	5.41%	3.60%
100%	5.41%	2.70%	1.80%

Table A2. Illustrative News Corp–OpenAI threshold sensitivity. This is not a transaction PIU estimate because the public record does not disclose cash-equivalent consideration, contemporaneous positive contribution margin, or the relevant product exposure.

The case is informative precisely because the standardized mature-company shortcut fails. A defensible valuation would need a forward revenue path, product contribution margin, expected useful life, the cash-equivalent value of credits, and possibly a strategic or option-value term. The framework exposes those missing primitives rather than hiding them in a single multiple.

A.4 The New York Times → Amazon, 2025

Transaction evidence. Reuters reported that the New York Times Company entered its first generative-AI licensing agreement with Amazon in May 2025. The multi-year agreement covers The New York Times, NYT Cooking, and The Athletic, permits real-time summaries and excerpts in Amazon products such as Alexa, and permits training of Amazon’s proprietary foundation models; the parties did not publicly disclose price [A6]. The Wall Street Journal subsequently reported annual consideration of approximately \$20 million–\$25 million and described the arrangement as cash-only [A7].

Buyer baseline. Amazon’s FY2024 Form 10-K reports consolidated net sales of \$637.959 billion and operating income of \$68.593 billion. AWS separately reported \$107.556 billion of sales and \$39.834 billion of operating income [A8]. At the midpoint reported annual fee of \$22.5 million and 5% consolidated scope,

$$\text{PIU}_{5\%, \text{mid}} = \frac{\$22.5\text{M}}{0.05 \times \$68.593\text{B}} = 0.656\%.$$

The reported \$20 million–\$25 million range maps to 0.583%–0.729% under the same 5% scope.

Standardized Amazon scope	Scoped FY2024 operating income	PIU at \$22.5M/year
1%	\$0.686B	3.280%
2.5%	\$1.715B	1.312%
5%	\$3.430B	0.656%
10%	\$6.859B	0.328%

Table A3. New York Times–Amazon recurring-license PIU sensitivity using the reported fee midpoint.

The business-unit choice illustrates model risk. If the same midpoint fee is instead compared with 5% of FY2024 AWS operating income, the threshold is

$$\frac{\$22.5\text{M}}{0.05 \times \$39.834\text{B}} = 1.130\%.$$

Neither basis is asserted to be the correct exposure. The agreement covers Alexa and foundation-model training, so a buyer-side appraisal would need to map the relevant economic base across those uses rather than mechanically choose consolidated Amazon or AWS.

A.5 News Corp → Meta, 2026

Transaction evidence. Meta publicly announced News Corp as a content partner for Meta AI in March 2026, describing broader real-time content and links from partner publications [A9]. The Wall Street Journal reported that the multi-year agreement could pay News Corp up to \$50 million per year for at least three years, with rights to retrieve current U.S. and U.K. information and train on other content such as archives [A10]. Because the number is an “up to” amount, it is a cap in the public reporting rather than an exact annual payment.

Buyer baseline. Meta’s FY2025 Form 10-K reports revenue of \$200.966 billion and income from operations of \$83.276 billion, an operating margin of approximately 41.44% [A11]. At the reported annual cap and 5% standardized scope,

$$\text{PIU}_{5\%} \leq \frac{\$50\text{M}}{0.05 \times \$83.276\text{B}} = 1.201\%.$$

The inequality matters: if realized annual consideration is below \$50 million, the corresponding break-even threshold is lower.

Standardized Meta scope	Scoped FY2025 operating income	PIU at \$50M/year cap
1%	\$0.833B	≤6.004%
2.5%	\$2.082B	≤2.402%
5%	\$4.164B	≤1.201%
10%	\$8.328B	≤0.600%

Table A4. News Corp–Meta upper-bound recurring-license PIU sensitivity.

A.6 Spirit as the one-time package benchmark

Spirit remains the paper’s most information-rich transaction because the public bankruptcy record describes the asset schedule, selected bid, alternate bid, deidentification obligation, and combined data/software package. The main text uses Google Cloud rather than consolidated Alphabet because the public evidence provides a plausible commercial deployment surface and segment economics. Under the 5% Google Cloud revenue-scope assumption, 35.6% segment operating-margin proxy, three-year life, 12% discount rate, $C = 0$, and $\rho = 1$, the selected \$10 million package bid implies PIU of approximately 0.236%. The 1%–10% scope sensitivity spans approximately 1.181% to 0.118%. As discussed in Sections 5, 6, and 8, this remains a package-level sensitivity because the filing does not allocate the bid between software and data and Google bears deidentification costs beyond the headline bid.

A.7 What the transaction panel does and does not validate

The cases provide three kinds of external support. First, they establish that AI-data/content consideration in the tens of millions of dollars per year is observable in actual contracting activity. Reddit’s securities filing additionally confirms a \$203 million aggregate licensing book with two- to

three-year terms at the moment Reuters reported the Google deal. Wiley, although not used for buyer-specific PIU because counterparties are not identified, has separately disclosed recurring AI licensing revenue and individual AI projects in public filings, further supporting the existence of a material market [A13].

Second, the cases demonstrate that Equation 3 can be run on heterogeneous transaction structures without hiding the assumptions. Recurring licenses yield a simple annual form when fee and benefit share the same timing. One-time acquisitions retain the discounted one-time formula. Caps produce upper-bound PIUs. Private loss-making buyers produce no defensible current-margin point estimate and therefore require a forward model.

Third, the panel does *not* establish that any licensed corpus generated the required uplift. No case in the public set supplies the controlled treatment/control data needed to measure Capability Alpha. A commercial license can be rational because of training value, retrieval/display value, legal certainty, product differentiation, strategic option value, or several channels at once. The empirical claim supported here is narrower: PIU is operational as a reverse-calculation instrument across observed prices, and it identifies the economic threshold that a subsequent data assay or production evaluation would need to clear.

Comparability remains limited. The contracts bundle different rights: static archives versus continuously updated streams; training versus retrieval and display; current versus historical content; cash versus credits; and possibly expertise, attribution, or other services. Reported consideration may be approximate and confidentiality prevents verification of many contractual details. The standardized 5% buyer-wide scope is intentionally mechanical and should not be mistaken for buyer-specific diligence. Finally, consolidated operating margin is a proxy, not an observed incremental margin for the affected AI product.

A.8 Additional source notes for the transaction panel

[A1] Reuters, “Reddit in AI content licensing deal with Google, sources say,” Feb. 22, 2024. Reports approximately \$60M annual consideration and use of Reddit content to train Google AI models. <https://www.reuters.com/technology/reddit-ai-content-licensing-deal-with-google-sources-say-2024-02-22/>

[A2] Reddit, Inc., Form S-1, filed Feb. 22, 2024. Discloses January 2024 data-licensing arrangements with \$203.0M aggregate contract value, two- to three-year terms, at least \$66.4M expected 2024 revenue, and concentration in one partner. <https://www.sec.gov/Archives/edgar/data/1713445/000162828024006294/reddits-1q423.htm>

[A3] Alphabet Inc., Form 10-K for FY2023. Revenue \$307.394B; operating income \$84.293B. <https://www.sec.gov/Archives/edgar/data/1652044/000165204424000022/goog-20231231.htm>

[A4] News Corp and OpenAI, “News Corp and OpenAI Sign Landmark Multi-Year Global Partnership,” May 22, 2024. Describes current/archive content access, display in responses, and product enhancement. <https://investors.newscorp.com/node/15946/pdf>

[A4b] The Wall Street Journal, “OpenAI, WSJ Owner News Corp Strike Content Deal Valued at Over \$250 Million,” May 22, 2024. Reports value above \$250M over five years, including cash and OpenAI credits, and training/query uses. <https://www.wsj.com/business/media/openai-news-corp-strike-deal-23f186ba>

[A5] Reuters, “Sam Altman’s OpenAI signs content agreement with News Corp,” May 22, 2024. Reports that WSJ valued the deal above \$250M over five years. <https://www.reuters.com/technology/sam-altmans-openai-signs-content-agreement-with-news-corp-2024-05-22/>

[A5b] Reuters, “OpenAI builds first chip with Broadcom and TSMC, scales back foundry ambition,” Oct. 29, 2024. Reports projected 2024 revenue of approximately \$3.7B and a projected \$5B loss. <https://www.reuters.com/technology/artificial-intelligence/openai-builds-first-chip-with-broadcom-tsmc-scales-back-foundry-ambition-2024-10-29/>

[A6] Reuters, “New York Times partners with Amazon for first AI licensing deal,” May 29, 2025. Describes a multi-year agreement covering Times, Cooking, and Athletic content, Alexa summaries/excerpts, and training Amazon proprietary foundation models; price undisclosed by parties. <https://www.reuters.com/business/retail-consumer/new-york-times-amazon-sign-ai-licensing-deal-2025-05-29/>

[A7] The Wall Street Journal, “Amazon to Pay New York Times at Least \$20 Million a Year in AI Deal.” Reports approximately \$20M–\$25M annual, cash-only consideration. <https://www.wsj.com/business/media/amazon-to-pay-new-york-times-at-least-20-million-a-year-in-ai-deal-66db8503>

[A8] Amazon.com, Inc., Form 10-K for FY2024. Consolidated net sales \$637.959B and operating income \$68.593B; AWS sales \$107.556B and operating income \$39.834B. <https://www.sec.gov/Archives/edgar/data/1018724/000101872425000004/amzn-20241231.htm>

[A9] Meta, “Bringing More International News and Content to Meta AI,” Mar. 13, 2026. Names News Corp as a Meta AI content partner and describes real-time content integration. <https://about.fb.com/news/2026/03/bringing-more-international-news-and-content-to-meta-ai/>

[A10] The Wall Street Journal, “News Corp, Meta in AI Content Licensing Deal Worth Up to \$50 Million a Year,” Mar. 3, 2026. Reports up to \$50M annually for at least three years and current/archive AI uses. <https://www.wsj.com/business/media/news-corp-meta-in-ai-content-licensing-deal-worth-up-to-50-million-a-year-d4fbf244>

[A11] Meta Platforms, Inc., Form 10-K for FY2025. Revenue \$200.966B; income from operations \$83.276B. <https://www.sec.gov/Archives/edgar/data/1326801/000162828026003942/meta-20251231.htm>

[A12] “Copyright and AI in the UK: Opting-In or Opting-Out?,” *GRUR International* 74(11), 2025. Reviews 83 known commercial model-training/data-access agreements and documents limited public detail on contractual terms. <https://academic.oup.com/grurint/article/74/11/1055/8209765>

[A13] John Wiley & Sons, FY2026/Q1 earnings materials and SEC filings. Wiley disclosed a \$20M AI licensing project with a large technology company, \$29M of AI revenue in Q1 FY2026, and \$92M lifetime AI revenue at that point; later FY2026 materials reported lifetime AI revenue above \$110M. <https://www.sec.gov/Archives/edgar/data/107140/000010714025000152/q126earningspresentation.htm>

Accessed August 24, 2026. Press-reported transaction values are used only as reported market signals and are not represented as contractual disclosures by the parties.